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PROB1:

```
import java.io.*;

public class p1 {
    public static void main(String[] args) {
        String inputFile = "./assignment8/src/input.txt";
        String outputFile = "./assignment8/src/output.txt";

        try {
            BufferedReader reader = new BufferedReader(new
            FileReader(inputFile));
            BufferedWriter writer = new BufferedWriter(new
            FileWriter(outputFile));

            String line;
            while ((line = reader.readLine()) != null) {
                int decimalNumber = Integer.parseInt(line);
                String binaryNumber = Integer.toBinaryString(decimalNumber);
                writer.write("Binary of " + decimalNumber + " is: " +
                binaryNumber);
                writer.newLine();
            }

            reader.close();
            writer.close();
            System.out.println("Conversion completed. Check the output file for
            results.");
        } catch (IOException e) {
            System.out.println("An error occurred: " + e.getMessage());
        } catch (NumberFormatException e) {
            System.out.println("Invalid number format: " + e.getMessage());
        }
    }
}
```

INPUT TEXT FILE:

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25
```

OUTPUT TEXT FILE:

```
Binary of 1 is: 1
Binary of 2 is: 10
Binary of 3 is: 11
Binary of 4 is: 100
Binary of 5 is: 101
Binary of 6 is: 110
Binary of 7 is: 111
Binary of 8 is: 1000
Binary of 9 is: 1001
Binary of 10 is: 1010
Binary of 11 is: 1011
Binary of 12 is: 1100
Binary of 13 is: 1101
Binary of 14 is: 1110
Binary of 15 is: 1111
Binary of 16 is: 10000
Binary of 17 is: 10001
Binary of 18 is: 10010
Binary of 19 is: 10011
Binary of 20 is: 10100
Binary of 21 is: 10101
Binary of 22 is: 10110
Binary of 23 is: 10111
Binary of 24 is: 11000
Binary of 25 is: 11001
```

PROB:2

```
import java.io.*;
import java.util.*;
public class p2 {
    public static void main(String[] args) {

        String inputFile = "./assignment8/src/input2.txt";
        String outputFile = "./assignment8/src/output2.txt";
        try {

            BufferedReader reader = new BufferedReader(new
FileReader(inputFile));
            StringBuilder paragraph = new StringBuilder();
            String line;

            while ((line = reader.readLine()) != null) {
                paragraph.append(line).append(" ");
            }
            reader.close();
            String[] words = paragraph.toString().toLowerCase().split("\\W+");

            Set<String> wordSet = new TreeSet<>(Arrays.asList(words));

            BufferedWriter writer = new BufferedWriter(new
FileWriter(outputFile));
            for (String word : wordSet) {
                writer.write(word);
                writer.newLine();
            }
            writer.close();
            System.out.println("Tokens have been successfully written to the
output file.");
        } catch (IOException e) {
            System.out.println("An error occurred: " + e.getMessage());
        }
    }
}
```

INPUT TEXT FILE:

```
1 Java is a high-level programming language. It was developed by Sun Microsystems in the mid-1990s.  
2 Java is known for its portability across platforms.  
3 This means that Java programs can run on any device that has the Java Virtual Machine installed.
```

OUTPUT TEXT FILE:

1	1990s
2	<u>a</u>
3	across
4	any
5	by
6	can
7	developed
8	device
9	for
10	has
11	high
12	in
13	installed
14	is
15	it
16	its
17	java
18	known
19	language
20	level
21	machine
22	means
23	microsystems
24	<u>mid</u>
25	<u>on</u>
26	platforms
27	portability
28	<u>programming</u>
29	<u>programs</u>
30	run
31	sun
32	that

PROB:3

```
import java.io.*;
import java.util.*;

class StudentDetails {
    protected String name;
    protected String regNo;
    protected String fatherName;

    public StudentDetails(String name, String regNo, String fatherName) {
        this.name = name;
        this.regNo = regNo;
        this.fatherName = fatherName;
    }
}

class Address extends StudentDetails {
    private String currentAddress;
    private String permanentAddress;
    private String contactNumber;
    private String email;

    public Address(String name, String regNo, String fatherName, String
currentAddress,
                    String permanentAddress, String contactNumber, String email)
{
    super(name, regNo, fatherName);
    this.currentAddress = currentAddress;
    this.permanentAddress = permanentAddress;
    this.contactNumber = contactNumber;
    this.email = email;
}

    public String getDetails() {
        return "Name: " + name + "\n"
            + "Registration Number: " + regNo + "\n"
            + "Father's Name: " + fatherName + "\n"
            + "Current Address: " + currentAddress + "\n"
            + "Permanent Address: " + permanentAddress + "\n"
            + "Contact Number: " + contactNumber + "\n"
            + "Email: " + email + "\n";
    }
}

public class p3 {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    List<Address> students = new ArrayList<>();
    String outputFile = "./assignment8/src/students.txt";
    try {
        System.out.println("Enter the number of students: ");
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            System.out.println("Enter details for student " + (i + 1) +
":");

            System.out.print("Name: ");
            String name = sc.nextLine();
            System.out.print("Registration Number: ");
            String regNo = sc.nextLine();
            System.out.print("Father's Name: ");
            String fatherName = sc.nextLine();
            System.out.print("Current Address: ");
            String currentAddress = sc.nextLine();
            System.out.print("Permanent Address:");
            String permanentAddress = sc.nextLine();
            System.out.print("Contact Number: ");
            String contactNumber = sc.nextLine();
            System.out.print("Email: ");
            String email = sc.nextLine();

            Address student = new Address(name, regNo, fatherName,
currentAddress, permanentAddress, contactNumber, email);
            students.add(student);
        }

        BufferedWriter writer = new BufferedWriter(new
FileWriter(outputFile));
        for (Address student : students) {
            writer.write(student.getDetails());
            writer.write("-----\n");
        }
        writer.close();
        System.out.println("Student details have been written to the output
file.");

        System.out.print("Enter the registration number to search: ");
        String searchRegNo = sc.nextLine();
        boolean found = false;
        for (Address student : students) {
            if (student.regNo.equals(searchRegNo)) {
                System.out.println("\nStudent details found:\n");
                System.out.println(student.getDetails());
                found = true;
            }
        }
    }
}

```

```

        break;
    }
}
if (!found) {
    System.out.println("The student with registration number " +
searchRegNo + " is not available in the record.");
}
} catch (IOException e) {
    System.out.println("An error occurred: " + e.getMessage());
}
}
}
}

```

student.txt:

```

1      Name: kiran
2      Registration Number: 20233093
3      Father's Name: parvathalu
4      Current Address: achampet
5      Permanent Address: gatthuthumman
6      Contact Number: 9653214756
7      Email: akk2005@gmail.com
8      -----
9      Name: maharshi
10     Registration Number: 20233059
11     Father's Name: ramu
12     Current Address: vishakapatnam
13     Permanent Address: vishakapatnam
14     Contact Number: 9654123807
15     Email: mahi98635@gmail.com
16     -----
17

```

PROB:4

```

import java.util.Scanner;

class TestScores {
    private double[] scores;

    public TestScores(double[] scores) throws IllegalArgumentException {
        this.scores = scores;
        for (double score : scores) {
            if (score < 0 || score > 100) {
                throw new IllegalArgumentException("Invalid test score: " +
score + ". Test scores should be between 0 and 100.");
            }
        }
    }

    public double getAverage() {
        double sum = 0;
        for (double score : scores) {
            sum += score;
        }
        return sum / scores.length;
    }
}

public class p4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of test scores: ");
        int numScores = sc.nextInt();
        double[] scores = new double[numScores];

        for (int i = 0; i < numScores; i++) {
            System.out.print("Enter score " + (i + 1) + ": ");
            scores[i] = sc.nextDouble();
        }

        try {
            TestScores testScores = new TestScores(scores);
            System.out.println("The average of the test scores is: " +
testScores.getAverage());
        } catch (IllegalArgumentException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}

```

OUTPUT:

```
"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\lib\idea_rt.jar=58711:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\bin" -jar C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\bin\idea_rt.jar 58711
Enter the number of test scores: 5
Enter score 1: 100
Enter score 2: 98
Enter score 3: 97
Enter score 4: 96
Enter score 5: 99
The average of the test scores is: 98.0
Process finished with exit code 0
```

PROB:5

```
import java.util.Scanner;

class InvalidTestScore extends Exception {
    public InvalidTestScore(String message) {
        super(message);
    }
}

class Scores {
    private double[] scores;

    public Scores(double[] scores) throws InvalidTestScore {
        this.scores = scores;
        for (double score : scores) {
            if (score < 0 || score > 100) {
                throw new InvalidTestScore("Invalid test score: " + score + ".
Test scores should be between 0 and 100.");
            }
        }
    }

    public double getAverage() {
        double sum = 0;
        for (double score : scores) {
            sum += score;
        }
        return sum / scores.length;
    }
}

public class p5 {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the number of test scores: ");
    int numScores = sc.nextInt();
    double[] scores = new double[numScores];

    for (int i = 0; i < numScores; i++) {
        System.out.print("Enter score " + (i + 1) + ": ");
        scores[i] = sc.nextDouble();
    }

    try {
        Scores testScores = new Scores(scores);
        System.out.println("The average of the test scores is: " +
testScores.getAverage());
    } catch (InvalidTestScore e) {
        System.out.println("Error: " + e.getMessage());
    }
}

```

OUTPUT:

```

"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\lib\idea_rt.jar=58737:C:\Program Files\JetBrains\Int
Enter the number of test scores: 5
Enter score 1: 100
Enter score 2: 98
Enter score 3: 97
Enter score 4: 96
Enter score 5: 99
The average of the test scores is: 98.0

Process finished with exit code 0

```

PROB:6

```

class InvalidPriceException extends Exception {
    public InvalidPriceException(String message) {
        super(message);
    }
}

class InvalidUnitsOnHandException extends Exception {
    public InvalidUnitsOnHandException(String message) {
        super(message);
    }
}

```

```

class RetailItem {
    private String description;
    private int unitsOnHand;
    private double price;

    public RetailItem(String description, int unitsOnHand, double price)
        throws InvalidUnitsOnHandException, InvalidPriceException {
        setDescription(description);
        setUnitsOnHand(unitsOnHand);
        setPrice(price);
    }

    public String getDescription() {
        return description;
    }

    public int getUnitsOnHand() {
        return unitsOnHand;
    }

    public double getPrice() {
        return price;
    }

    public void setDescription(String description) {
        this.description = description;
    }

    public void setUnitsOnHand(int unitsOnHand) throws
InvalidUnitsOnHandException {
        if (unitsOnHand < 0) {
            throw new InvalidUnitsOnHandException("Units on hand cannot be
negative.");
        }
        this.unitsOnHand = unitsOnHand;
    }

    public void setPrice(double price) throws InvalidPriceException {
        if (price < 0) {
            throw new InvalidPriceException("Price cannot be negative.");
        }
        this.price = price;
    }
}

public class p6 {
    public static void main(String[] args) {
        try {

```

```

        RetailItem item1 = new RetailItem("Jacket", 12, 59.95);
        RetailItem item2 = new RetailItem("Designer Jeans", 40, 34.95);
        RetailItem item3 = new RetailItem("Shirt", 20, 24.95);

        displayItem(item1);
        displayItem(item2);
        displayItem(item3);

        System.out.println("\nTrying to create an item with negative
price:");
        RetailItem invalidItem = new RetailItem("Invalid Item", 5, -9.99);
    } catch (InvalidPriceException e) {
        System.out.println("Error: " + e.getMessage());
    } catch (InvalidUnitsOnHandException e) {
        System.out.println("Error: " + e.getMessage());
    }
}

public static void displayItem(RetailItem item) {
    System.out.println("Description: " + item.getDescription());
    System.out.println("Units on Hand: " + item.getUnitsOnHand());
    System.out.println("Price: " + item.getPrice());
    System.out.println("-----");
}
}

```

OUTPUT:

```

Description: Jacket
Units on Hand: 12
Price: 59.95
-----
Description: Designer Jeans
Units on Hand: 40
Price: 34.95
-----
Description: Shirt
Units on Hand: 20
Price: 24.95
-----

```

PROB:7

```
import java.io.*;

class FileArray {
    public static void writeArray(String filename, int[] array) throws
IOException {
        try (DataOutputStream out = new DataOutputStream(new
FileOutputStream(filename))) {
            for (int i : array) {
                out.writeInt(i);
            }
        }
    }

    public static void readArray(String filename, int[] array) throws
IOException {
        try (DataInputStream in = new DataInputStream(new
FileInputStream(filename))) {
            for (int i = 0; i < array.length; i++) {
                array[i] = in.readInt();
            }
        }
    }
}

public class p7 {
    public static void main(String[] args) {
        int[] writeArray = {10, 20, 30, 40, 50};
        String filename = "arrayData.bin";

        try {
            FileArray.writeArray(filename, writeArray);
            System.out.println("Array successfully written to the file.");
        } catch (IOException e) {
            System.out.println("Error writing to file: " + e.getMessage());
        }

        int[] readArray = new int[writeArray.length];

        try {
            FileArray.readArray(filename, readArray);
            System.out.println("Array successfully read from the file.");
        } catch (IOException e) {
            System.out.println("Error reading from file: " + e.getMessage());
        }

        System.out.println("Contents of the read array:");
        for (int i : readArray) {
```

```

        System.out.print(i + " ");
    }
}

```

OUTPUT:

```

"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\lib\idea_rt.jar=58574:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\bin\idea_rt.jar" "-Didea.config.path=C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\config" "-Didea.system.path=C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.1.3\lib" "-Didea.version=2024.1.3" "Main.java"
Array successfully written to the file.
Array successfully read from the file.
Contents of the read array:
10 20 30 40 50
Process finished with exit code 0

```

PROB:8

```

import java.io.*;

class FileEncryptionFilter {
    public static void encryptFile(String inputFile, String outputFile) throws
IOException {
        try (FileInputStream inFile = new FileInputStream(inputFile);
             FileOutputStream outFile = new FileOutputStream(outputFile)) {
            int data;
            while ((data = inFile.read()) != -1) {
                int encryptedData = data + 10;
                outFile.write(encryptedData);
            }
        }
    }
}

public class p8 {
    public static void main(String[] args) {
        String inputFile = "./assignment8/src/plaintext.txt";
        String outputFile = "./assignment8/src/encryptedtext.txt";
        try {
            FileEncryptionFilter.encryptFile(inputFile, outputFile);
            System.out.println("File has been successfully encrypted!");
        } catch (IOException e) {
            System.out.println("Error occurred: " + e.getMessage());
        }
    }
}

```

```
}
```

INPUT TEXT FILE:

```
This is a secret message from the universe. A silent manifestation of yours is soon going to  
come true.
```

OUTPUT TEXT FILE:

```
rs}*s}*k*}om|o~*wo}}kqo*p|yw*~ro*DELxs0o|}o8*K*}svox~*wkxspo}~k~syx*yp*0yDEL|}*s}*}yyx*qysxq*~yETBDC4mywo*~|DELo8
```

PROB:9

```
import java.io.*;

class FileDecryptionFilter {
    public static void decryptFile(String inputFile, String outputFile) throws
IOException {
        try (FileInputStream inFile = new FileInputStream(inputFile);
             FileOutputStream outFile = new FileOutputStream(outputFile)) {
            int data;
            while ((data = inFile.read()) != -1) {
                int decryptedData = data - 10;
                outFile.write(decryptedData);
            }
        }
    }
}

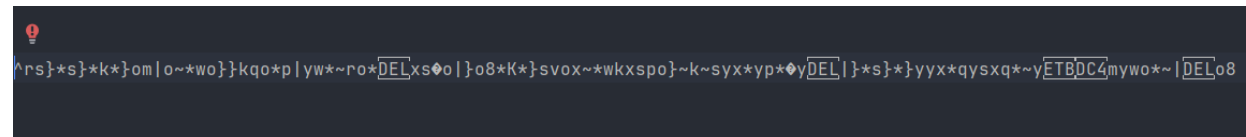
public class p9 {
    public static void main(String[] args) {
        String inputFile = "./assignment8/src/encryptedText.txt";
        String outputFile = "./assignment8/src/decryptedText.txt";
        try {
```

```

        FileDecryptionFilter.decryptFile(inputFile, outputFile);
        System.out.println("File has been successfully decrypted!");
    } catch (IOException e) {
        System.out.println("Error occurred: " + e.getMessage());
    }
}
}

```

INPUT TEXT FILE:

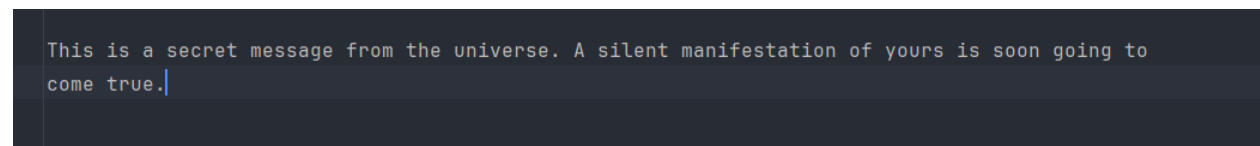


```

^rs}*s}*k*}om|o~*wo}}kqo*p|yw*~ro*DELxs0|}o8*K*}svox~*wkxspo}~k~syx*yp*0yDEL|}*s}*}yyx*qysxq*~yETBDC4mywo*~|DELo8

```

OUTPUT TEXT FILE:



```

This is a secret message from the universe. A silent manifestation of yours is soon going to
come true.

```